

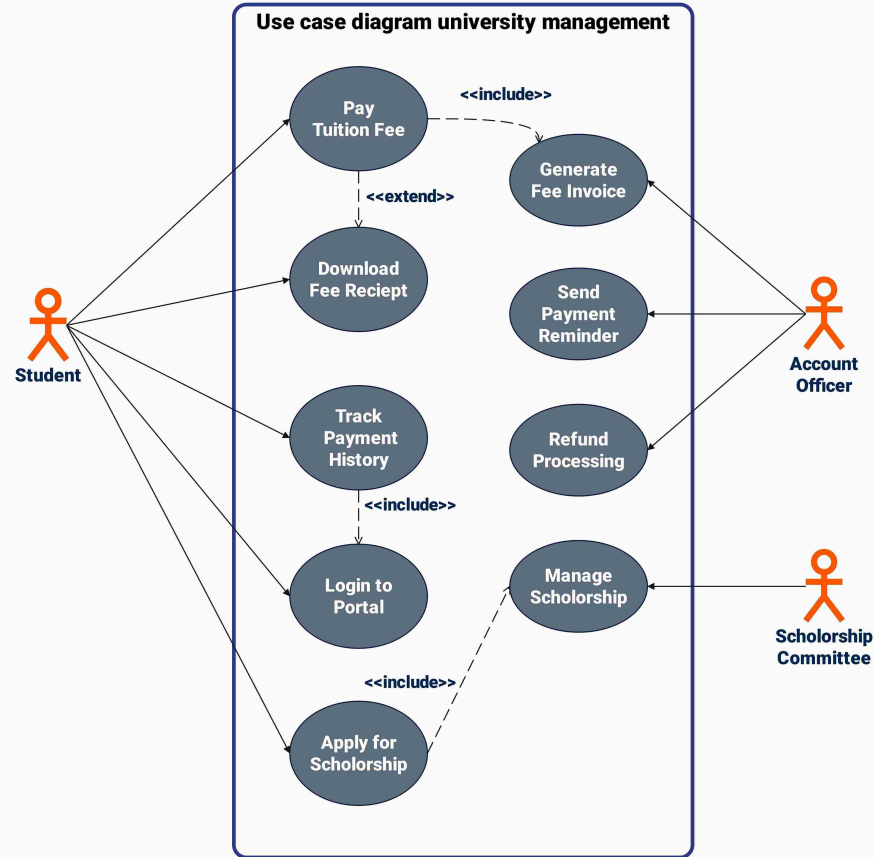
Use Case Models

Use Case Diagrams

A use case model is a visual representation of the interaction between users (actors) and the system to achieve specific goals (use cases).

It defines **what** the system should do from the user's perspective, not how the system will do it

Example



Terms

- **Actor:**

An entity that *interacts* with the system (e.g., user, *external system*).

- **Use Case:**

A *specific function* or *action* that the system performs in response to an actor's request.

- **System Boundary:**

A box that defines the *scope* of the system and separates it from *external actors*.

Actor

An actor represents a role that interacts with the system. It can be a *person*, another *system*, or an *organization*.

Use Case

A use case represents a *specific function* or *action* that the system performs in response to an actor's request.

It describes a *sequence of interactions* between the actor and the system to achieve a particular *goal*.

System Boundary

The system boundary is a box that defines the scope of the system and **separates** it from external actors.

Benefits

1. **Clarity**: Use case diagrams provide a clear and *visual* representation of the system's functionality and its interactions with users.
2. **Communication**: They facilitate communication between stakeholders, developers, and designers by providing a *common language* to discuss system requirements.
3. **Requirement Analysis**: Use case diagrams help in identifying and analyzing the functional requirements of the system, ensuring that all necessary features are captured and understood.

Note that any form of documentation, as long as it's thorough enough will help in indentifying requirements

A note on UML

Use case diagrams are part of the *Unified Modeling Language* (UML), a standardized modeling language used in software engineering to visualize the design of a system.

One of the primary benefits of UML is that it provides a *common language for developers, designers, and stakeholders* to communicate and understand the structure and behavior of a system.

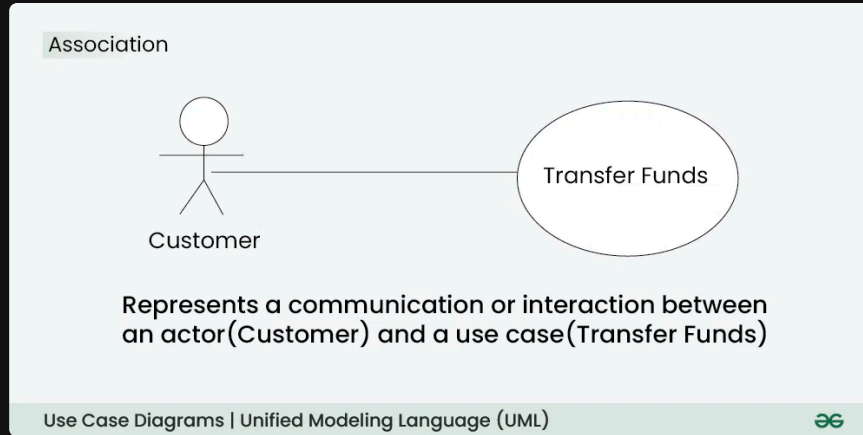
It was made in the 1990s by Grady Booch, Ivar Jacobson, and James Rumbaugh, who were working at Rational Software Corporation.

They created UML to *unify the various modeling languages* that were being used at the time and to provide a standard way to visualize software design.

Relationships

1. **Association**: A relationship between an actor and a use case, indicating that the actor participates in the use case.
2. **Include**: A relationship where one use case includes the behavior of another use case
3. **Extend**: A relationship where one use case extends the behavior of another use case under certain conditions.
4. **Generalization**: A relationship where one actor or use case is a specialized version

Association

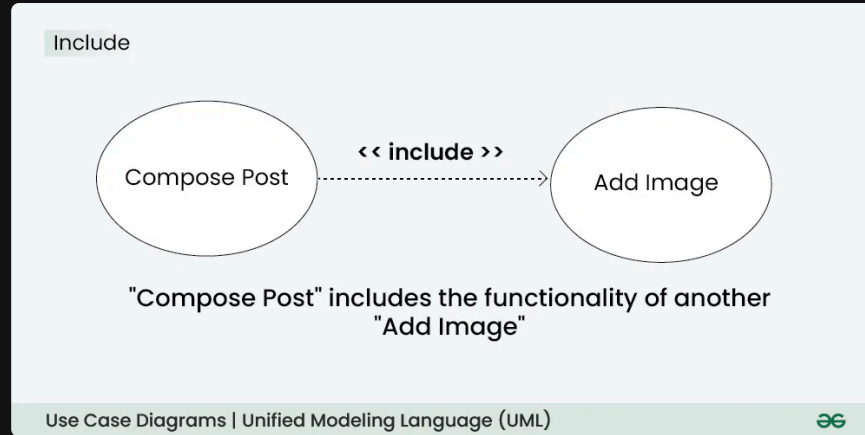


Represents an *interaction* between an actor and a use case.

It's depicted by a line, and is used to show a **relationship**

Does not necessarily indicate a *direct* interaction, but rather that the actor is involved in the use case in some way.

Include

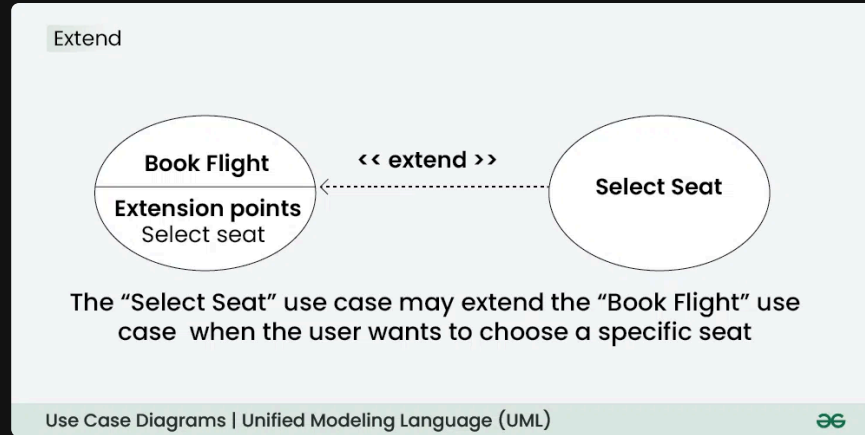


Represents a use case that *includes* the behavior of *another use case*.

It's depicted by a dashed arrow with the label "include", and is used to show that one use case is a *subroutine* of another use case.

It's primarily used for *modularity*

Extend



Represents that a use case can be *extended* by another use case under certain conditions.

It's depicted by a dashed arrow with the label "extend", and is used to show that one use case can *add* behavior to another use case.

Mostly used for optional behavior that is not always executed, but can be executed under certain conditions.

Generalization

An "**is-a**" relationship between actors or use cases, where one actor or use case is a *specialized version* of another.

Depicted by a *solid line with a hollow arrowhead* pointing from the specialized actor or use case to the more general actor or use case.

For example, a *vehicle renting system* having a "rent car" and "rent bike" which is a specialized version of "rent vehicle"

Activity: Use case diagram

Choose one business scenario to create a Use Case Diagram for on the next slide
it must

- have at least *two actors*
- define at least *3-5 core* functionalities as use cases
- include a clear *system boundary*
- use *association* relationships
- must include at least one "*include*" or "*extend*" relationship

use draw.io

title it `lastname_firstname_usecase.png` and submit it to neo

Activity: Use case diagram

1. **Local Bakery:** Automate customer order management and track real-time ingredient inventory.
2. **Pet Grooming Salon:** Manage appointment scheduling and maintain detailed customer pet profiles.
3. **Small Gym/Fitness Studio:** Track member monthly subscriptions and handle class bookings.
4. **Independent Bookstore:** Manage a digital book catalog and track sales/stock levels.
5. **Coffee Shop:** Streamline point-of-sale (POS) transactions and track customer loyalty rewards.
6. **Pharmacy:** Manage patient prescription records and automate medication refill alerts.
7. **Dry Cleaning Service:** Track garment processing status and notify customers when items are ready.
8. **Car Wash/Detailing:** Manage tiered service packages and schedule vehicle appointments.